

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of BSc (Biotechnology/Microbiology) Examination, March 2018

Course code & Name: BT501: Principles of Bioprocess engineering and Technology

Date: 28.03.2018 Day: Wednesday Time: 01.30 pm To 03.30 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	10
1.	In bioreactors, the role of an impeller is to provide (i) foam control (ii) mixing (iii) temperature control (iv) none of the above	
2.	Air to be supplied to the fermenter can be sterilized using (i) filtration (ii) heating (iii) UV rays (iv) all the above	
3.	Which of following is a product of plant cell culture? (i) penicillin (ii) monoclonal antibody (iii) ethanol (iv) Taxol	
4.	Which of the following statement(s) regarding immobilized enzyme is/are TRUE? (i) Component enzymes may be reused (ii) The cell density or enzyme density in a location can be increased (iii) Enzymes stability is improved (iv) All of the above	
5.	In a fermenter, air flow rate can be measured using: (i) thermometer (ii) rotameter (iii) pH meter (iv) tachometer	

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Date: 28.03.2018 Day: Wednesday Time: 01.30 pm To 03.30 pm Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed		10
Q-II (A)	Answer any one of the following: (i) Discuss the advantages of biological processes over chemical processes for production of useful compounds. (ii) Discuss the use of any one of the cheaper carbon source used in formulation of industrial scale fermentation medium.	04
Q-II (B)	Answer any two of the following: (i) Why immobilization of enzymes is required? Explain any one method for immobilization of enzymes. (ii) Write a note on: Continuous steam injection and flash cooling sterilization process. (iii) Discuss inoculum preparation for fermentation processes involving bacterial cultures.	06
SECTION – II		
Q-III Answer the following questions as directed		15
Q-III (A)	Answer any two of the following: (i) Give a labeled diagram for a fermenter and its parts. Explain the role of sparger in fermenter. (ii) Giving an example discuss the advantages and disadvantages of SSF (iii) What is the need for mixing in bioreactors? Give labelled diagram for any three types of agitators.	06

Q-III (B)	Answer any two of the following: (i) Write a note on: cultivation of plant cells. (ii) What is non-growth associated product formation kinetics? Give only diagram for fermentation profile of a secondary metabolite such as penicillin. (iii) Comment: Cost of medium influences economics of a fermentation process.	06
Q-III (C)	Giving its importance describe any one of the techniques used for downstream processing in a fermentation industry.	03